

PH1456 : Tut 2

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Today: Central Potentials, Partial Waves.

Last week: plane wave in $\hat{z}$ direction as a sum of spherical waves:

$$e^{ikz} = \sum_{l=0}^{\infty} i^l (2l+1) j_l(kr) P_l(\cos\theta)$$

IMPORTANT

$(l^{\text{th}} \text{ Legendre Polynomial})$
 $P_l(x) \equiv \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2-1)^l$
(take $x = \cos\theta$)

$j_l(x) \equiv (-1)^l x^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \left[\frac{\sin x}{x} \right]$
(Spherical Bessel function of first kind)

which can also be written in terms of spherical harmonics

$$Y_l^0(\theta) \equiv \sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta) \quad (\text{Spherical Harmonics of } 0^{\text{th}} \text{ order})$$

Next: Free TISE can be written in spherical coordinates.

* If you require the energy eigenfunctions $\psi^{(0)}(\vec{r})$ also be

simultaneous eigenfunctions of L^2, L_z , then
(orbital ang. momentum)

$$\varphi_{\text{ken}}^{(o)}(\vec{r}) = \sqrt{\frac{2\kappa^2}{\pi}} j_\ell(\kappa r) Y_\ell^m(\theta, \varphi).$$

It follows that

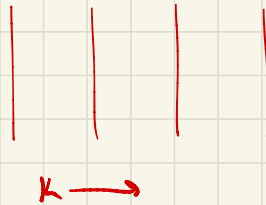
$$\hat{H}_0 \varphi_{\text{ken}}^{(o)}(\vec{r}) = \left(\frac{\hbar^2 \kappa^2}{2\mu} \right) \varphi_{\text{ken}}^{(o)}(\vec{r})$$

$$\hat{L}^2 \varphi_{\text{ken}}^{(o)}(\vec{r}) = [\ell(\ell+1) \hbar^2] \varphi_{\text{ken}}^{(o)}(\vec{r})$$

$$\hat{L}_z \varphi_{\text{ken}}^{(o)}(\vec{r}) = (m\hbar) \varphi_{\text{ken}}^{(o)}(\vec{r}).$$

Otherwise: you just get plane waves written in spherical coordinates:

1) Solutions to free TISE (in any coordinate basis system) : plane waves



2) Solutions to TISE where $[\hat{H}_0, \hat{L}^2] = [\hat{H}_0, \hat{L}_z] = [\hat{L}_z, \hat{L}^2] = 0$

in spherical coordinates:



these commutations define $\varphi_{\text{ken}}(\vec{r})$, which can easily be written in spherical coordinates although operators are coordinate-basis independent.

In scattering theory, we are interested in the $r \rightarrow \infty$ limit behavior of $\varphi_{k\ell m}^{(o)}(\vec{r})$:

$$* \lim_{r \rightarrow \infty} j_\ell(kr) = \frac{1}{kr} \sin(kr - \frac{\ell\pi}{2})$$

we have

$$\varphi_{k\ell m}^{(o)} \underset{r \rightarrow \infty}{\sim} c_\ell \frac{\sin(kr - \frac{\ell\pi}{2})}{r} Y_\ell^m(\theta, \varphi)$$

- $c_\ell = c_\ell(k)$ some function of k
- $r \rightarrow \infty$ spherical limit
- $|j_\ell(k, \varphi)|^2 \propto$ to a phase shift.

In general: Now take as Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}(r)$, $r = |\vec{r}|$. central

The general solutions that are also Eigenfunctions of $\hat{L}_z, \hat{L}^2$

take the form (Variable separation)

"Partial waves"

$$\varphi_{k\ell m}(\vec{r}) = R_{k\ell}(r) Y_\ell^m(\theta, \varphi)$$

↑ know what these are

Write / Substitute into radial equation $R_{k\ell}(r) = \frac{1}{r} u_{k\ell}(r)$:

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} - \underbrace{\frac{\ell(\ell+1)\hbar^2}{2\mu r^2} + V(r)}_{\text{EFFECTIVE POTENTIAL}} \right] u_{k\ell}(r) = \frac{\hbar^2 k^2}{2\mu} u_{k\ell}(r).$$

EFFECTIVE POTENTIAL
 $V_{\text{eff}}(r)$

(See VIII complement A)
↳ CC-T

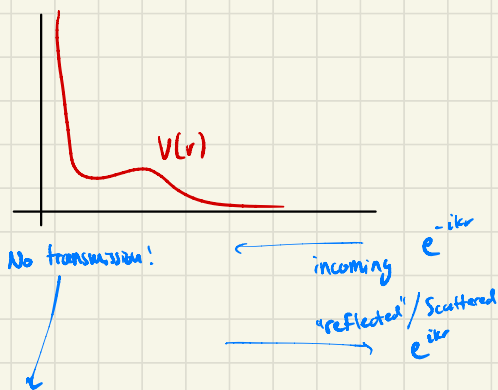
With the boundary condition $u_{k\ell}(r=0) = 0$ which must be true for normalization since $V_{\text{eff}}(r) \rightarrow \infty$ as $r \rightarrow 0$ in second term.
and $V(r) \rightarrow 0$ as $r \rightarrow \infty$ ALWAYS in scattering theory.

Then as $r \rightarrow \infty$, we have

$$\left[\frac{d^2}{dr^2} + k^2 \right] u_{k\ell}(r) \stackrel{(\text{limit})}{\approx} 0.$$

$$\Rightarrow u_{k\ell}(r) = A e^{ikr} + B e^{-ikr}$$

ie.



which implies $|A|^2 = |B|^2$:

$$u_{k\ell}(r) = |A| \left(e^{ikr} e^{i\phi_A} + e^{-ikr} e^{i\phi_B} \right) \quad (\phi_A, \phi_B \text{ phases})$$

$$\underset{r \rightarrow \infty}{\text{(equivalently)}} \sim C \cdot \sin \left(kr - \frac{r\pi}{2} + \delta_\ell \right) \quad \text{phase shift}$$

Thus

$$\psi_{k\ell m}(\vec{r}) = C \frac{\sin \left(kr - \frac{r\pi}{2} + \delta_\ell \right)}{r} Y_\ell^m(\theta, \varphi)$$

which is just $\psi_{k\ell m}^{(0)}$ with a phase shift δ_ℓ .

The total cross-section (entire scattering) is then

$$\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2(\delta_l)$$

Generated by
the phase shift!